UNIVERSITY OF WAH(UOW)

DEPARTMENT OF COMPUTER SCIENCE

Lab Task # 01

COURSE: Data Structure & Algorithm Lab_201L

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Task # 01

```cpp
int main() {
    string name, regNumber;
    int semester, yearJoined, currentYear = 2025, yearsStudied;
    cout << "Enter your name: ";
    cin >> name;
    cout << "Enter your registration number: ";
    cin >> regNumber;
    cout << "Enter your semester: ";
    cin >> semester;
    cout << "Enter the year you joined the university: ";
    cin >> yearJoined;
    yearsStudied = currentYear - yearJoined;
    cout << "\nName: " << name << endl;
    cout << "Registration Number: " << regNumber << endl;
    cout << "Semester: " << semester << endl;
    cout << "Year Joined: " << yearJoined << endl;
    cout << "You have studied for " << yearsStudied << " years." << endl;

    return 0;
}
```

Output

```
Enter your name: Talha
Enter your registration number: 101
Enter your semester: 3
Enter the year you joined the university: 1

Name: Talha
Registration Number: 101
Semester: 3
Year Joined: 1
You have studied for 2024 years.
```

Task # 02

```cpp
int main() {
    int a, b;
    cout << "Enter first number: ";
    cin >> a;
    cout << "Enter second number: ";
    cin >> b;
    cout << "\nSum: " << a + b << endl;
    cout << "Difference: " << a - b << endl;
    cout << "Product: " << a * b << endl;
    cout << "Quotient: " << a / b << endl;
    cout << "Remainder: " << a % b << endl;
    if (a > b)
        cout << a << " is greater than " << b << endl;
    else if (b > a)
        cout << b << " is greater than " << a << endl;
    else
        cout << "Both numbers are equal." << endl;
    return 0;
```

Output

```
Enter first number: 102
Enter second number: 123

Sum: 225
Difference: -21
Product: 12546
Quotient: 0
Remainder: 102
123 is greater than 102
```

Task # 03

```cpp
int main() {
    float width, height, area, perimeter;
    cout << "Enter width: ";
    cin >> width;
    cout << "Enter height: ";
    cin >> height;
    area = width * height;
    perimeter = 2 * (width + height);
    cout << "\nArea: " << area << endl;
    cout << "Perimeter: " << perimeter << endl;
    if (width == height)
        cout << "It is a square." << endl;
    else
        cout << "It is a rectangle." << endl;
```

Output

```
Enter width: 32.1
Enter height: 44.1

Area: 1415.61
Perimeter: 152.4
It is a rectangle.
```

Task # 04

```
int main() {
    float num1, num2, num3, average;
    cout << "Enter three numbers: ";
    cin >> num1 >> num2 >> num3;
    average = (num1 + num2 + num3) / 3;
    cout << "\nAverage: " << average << endl;
    if (average >= 50)
        cout << "Pass" << endl;
    else
        cout << "Fail" << endl;
    return 0;
```

Output

```
Enter three numbers: 33 55 66

Average: 51.3333
Pass
```

Task # 05

```
int main() {
    float mark1, mark2, mark3, total, average, percentage;
    cout << "Enter marks for three subjects: ";
    cin >> mark1 >> mark2 >> mark3;
    total = mark1 + mark2 + mark3;
    average = total / 3;
    percentage = (total / 300) * 100;
    cout << "\nTotal: " << total << endl;
    cout << "Average: " << average << endl;
    cout << "Percentage: " << percentage << "%" << endl;
    if (percentage >= 80)
        cout << "Grade: A" << endl;
    else if (percentage >= 70)
        cout << "Grade: B" << endl;
    else if (percentage >= 60)
        cout << "Grade: C" << endl;
    else if (percentage >= 50)
        cout << "Grade: D" << endl;
    else
        cout << "Grade: Fail" << endl;
```

Output

```
Enter marks for three subjects: 22 33 55

Total: 110
Average: 36.6667
Percentage: 36.6667%
Grade: Fail
```